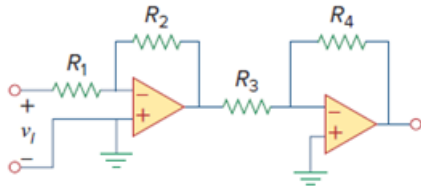


Problem

Using Fig. 5.83, design a problem to help other students better understand cascaded op amps.

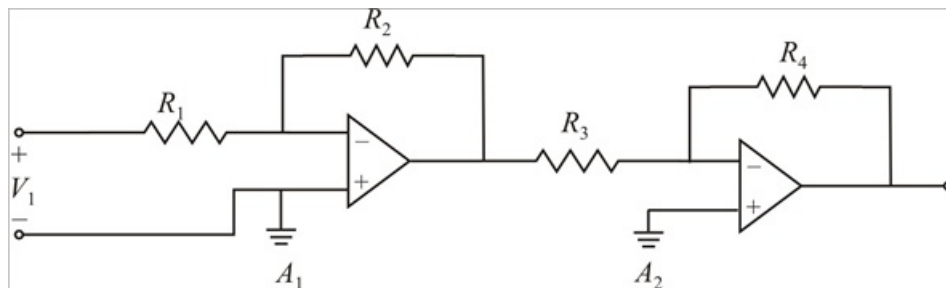
Figure. 5.83:



Step-by-step solution

Step 1 of 3

Considering the circuit



Step 2 of 3

The total gain of the above circuit is

$$A = A_1 \times A_2$$

Observing the circuit it is a cascading of two OP amps

Both the OP amp circuits are inverting amplifiers

Step 3 of 3

The gain of A_1 is

$$A = -\frac{R_2}{R_1}$$

The gain of A_2 is

$$A = -\frac{R_4}{R_3}$$

The total gain is

$$A = A_1 \times A_2$$

The total gain

$$A = \frac{V_0}{V_i}$$

$$\frac{V_0}{V_i} = -\frac{R_2}{R_1} \times -\frac{R_4}{R_3}$$

$$\frac{V_0}{V_i} = \frac{R_2 R_4}{R_1 R_3}$$

$$V_0 = \frac{R_2 R_4}{R_1 R_3} V_i$$